

What fills the context window

Every call re-sends the whole thing, and all of it competes for one ceiling.

COMPONENT

PROCESS

TECHNIQUE

CONTROL

CEILING

ARTIFACT

STORE

↑ THE WINDOW

a per-call ceiling on all of it together — not a budget that refills

01

TECHNIQUE

System prompt

the vendor's standing instructions

Role, rules and format, sent ahead of the conversation on every request. It is the part of the product you cannot read or edit.

YOU NEVER SEE IT

02

COMPONENT

Tool definitions

what it is allowed to ask for

Every function's name, description and arguments. They are part of the prompt, so a large toolset is a standing cost on the window.

CHARGED WHETHER USED OR NOT

03

STORE

The conversation

there is nowhere else for it to live

Not a store the agent writes to — the transcript, re-sent whole. This is the part that runs you out of room.

GROWS EVERY TURN

⚡ Compaction

Older turns replaced by a summary so a long run keeps fitting.

🔄 Session

Ends, and takes the window with it. A fresh one often beats arguing.

04

TECHNIQUE

Retrieved material

fetches at question time

Documents pulled in for this question, so the answer rests on them rather than on what the model absorbed in training.

THE POINT IS GROUNDING

⚡ Chunking

Documents split small enough to retrieve a piece rather than the whole.

⚡ Semantic search

Retrieval by meaning rather than by matching words.

⚡ Reranking

A second pass that reorders hits before any of them cost tokens.

🛡️ Grounding

Answering from the retrieved source instead of from memory.

THE UNIT IT IS MEASURED IN

📄 **Token** — A word fragment. The window, the bill and the rate limits are all counted in these, not in words.

THE WORK OF DECIDING

⚡ **Context engineering** — What goes in, what gets dropped, what gets fetched again — most of what people call prompt engineering is this.

WHAT IT IS NOT

📖 **Long-term memory** — State across sessions is a directory the harness reads back into the window. The window itself remembers nothing.

Nothing in the window persists. Every call re-sends all of it, the ceiling applies to the whole, and what looks like memory is a file something re-read for you.